Agent Generated Hypothesis and Research Plan

Novel Hypothesis:

Metformin, by modulating mitochondrial function and the AMPK/mTOR pathway, disrupts compensatory metabolic reprogramming in GBM cells, specifically inhibiting the upregulation of glutamine metabolism and fatty acid oxidation that occurs in response to glycolysis inhibition or other metabolic stresses. This metabolic disruption sensitizes GBM cells to standard therapies like temozolomide and radiation, leading to improved treatment outcomes. Furthermore, metformin can overcome resistance to therapies targeting glycolysis or mTOR, by inhibiting glutamine and fatty acid oxidation.

Research Plan:

This research plan aims to test the hypothesis that metformin can disrupt compensatory metabolic reprogramming in GBM cells, leading to improved treatment outcomes when combined with standard therapies.

- **Cell Culture Models:**
- Establish GBM cell lines (e.g., U87-MG, A172, LN229) that exhibit varying levels of glycolytic activity, glutamine dependence, and fatty acid oxidation. This can be assessed by measuring glucose consumption, lactate production, glutamine uptake, and fatty acid oxidation rates in each cell line.
- Develop glycolysis-resistant GBM cell lines by chronic exposure to glycolytic inhibitors like 2-deoxyglucose (2-DG) or by CRISPR/Cas9-mediated knockout of key glycolytic enzymes. These resistant cell lines will model GBM cells that have adapted to glycolytic stress by upregulating alternative metabolic pathways.
- **Metabolic Assays:**
- **Glucose Consumption and Lactate Production:** Measure glucose consumption and lactate production rates using commercially available kits to assess glycolytic activity.
- **Glutamine Uptake and Metabolism:** Quantify glutamine uptake and glutamate production using HPLC or mass spectrometry. Assess the expression of glutaminase (GLS) and other key enzymes involved in glutamine metabolism using Western blotting and qPCR.
- **Fatty Acid Oxidation (FAO) Assay:** Measure FAO rates using a radioactive or fluorescent substrate (e.g., palmitate) and quantify the production of CO2 or other metabolites. Assess the expression of carnitine palmitoyltransferase 1A (CPT1A) and other key FAO enzymes.
- **ATP Production:** Measure ATP levels using a luciferase-based assay to assess the impact of metformin on cellular energy production.
- **Mitochondrial Respiration:** Perform Seahorse XF analysis to measure oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) to assess mitochondrial function and glycolytic activity.
- **Molecular Analyses:**
- **Western Blotting:** Analyze the expression and phosphorylation status of key proteins in the AMPK/mTOR pathway (e.g., AMPK, mTOR, p70S6K, 4E-BP1) to assess the impact of metformin on this pathway. Also, assess expression changes in metabolic enzymes like Glutaminase, CPT1A, etc.
- **qRT-PCR:** Measure the mRNA expression levels of genes involved in glycolysis, glutamine metabolism, and fatty acid oxidation to assess the impact of metformin on gene expression.

- **Metabolomics:** Perform untargeted metabolomics analysis using LC-MS/MS to identify changes in metabolite levels in response to metformin treatment.
- **Combination Studies:**
- Treat GBM cell lines with metformin alone and in combination with temozolomide (TMZ) or radiation.
- Assess cell viability using MTT or CellTiter-Glo assays.
- Perform clonogenic assays to assess the long-term impact of the treatments on cell survival.
- Analyze cell cycle distribution using flow cytometry.
- Measure apoptosis using Annexin V/PI staining and flow cytometry.
- **In Vivo Studies:**
- Establish orthotopic GBM xenograft models by injecting GBM cells into the brains of immunocompromised mice.
- Treat mice with metformin alone and in combination with TMZ or radiation.
- Monitor tumor growth using MRI or bioluminescence imaging.
- Assess survival rates.
- Perform immunohistochemical analysis of tumor tissues to assess the expression of key metabolic enzymes, AMPK/mTOR pathway proteins, and proliferation markers (e.g., Ki-67).

Expected Outcomes and Data Analysis:

- We anticipate that metformin will reduce GBM cell viability, particularly in cells that have adapted to glycolysis inhibition by upregulating glutamine metabolism and/or fatty acid oxidation.
- Metformin will modulate the AMPK/mTOR pathway and alter the expression of key metabolic enzymes, leading to decreased ATP production and increased oxidative stress.
- The combination of metformin with TMZ or radiation will result in synergistic anti-tumor effects in vitro and in vivo.

Novelty Assessment:

Based on the PubMed searches conducted during this process, the core hypothesis appears novel, although there is some overlap with existing research.

- **Metformin and GBM:** The initial search revealed some studies on metformin and GBM, but these primarily focused on the Warburg effect (aerobic glycolysis) or the combination of metformin with other drugs.
- **Metabolic Reprogramming:** The concept of metabolic reprogramming in GBM is well-established, but the specific focus on *compensatory* upregulation of glutamine metabolism and fatty acid oxidation in response to glycolysis inhibition, and metformin's potential to disrupt this compensation, is less explored. PMID:25798620 supports the idea of compensatory glutamine metabolism in the context of mTOR inhibition.
- **Fatty Acid Oxidation:** While several studies highlight the importance of FAO in GBM, the novelty here lies in the combination with glycolysis inhibition and the potential for metformin to simultaneously target multiple metabolic pathways.
- **MET Inhibition and PGC1alpha:** PMID:31694905 highlights the link between MET inhibition and metabolic reprogramming in GBM, indicating that targeting receptor tyrosine kinases can influence cancer metabolism.

Therefore, while the individual components of the hypothesis (metformin, GBM, metabolic reprogramming, glutamine metabolism, fatty acid oxidation) have been investigated separately, the *specific combination of these elements, focusing on metformin's ability to disrupt compensatory metabolic adaptations to glycolysis inhibition by targeting glutamine metabolism and fatty acid oxidation,* represents a novel and testable hypothesis for repurposing metformin in GBM.